ONE-SIDED KAZHDAN TRANSPORT AND MF RADICALS

SAUERS

We study homomorphisms into norm matrix coronas. If a property-(T) subgroup is conjugated into itself, one-sided Kazhdan transport preserves its Hilbert–Schmidt asymptotic commutant. Consequently, commutators between the transported commutant and the subgroup are universally Hilbert–Schmidt trivial in operator-norm asymptotic representations. We prove that every normal property-(T) subgroup contained in the resulting defect subgroup lies in the MF radical. In particular, when the defect subgroup is the ambient Kazhdan group, every homomorphism to an MF group is trivial. For the binary Leavitt algebra this gives

$$\mathrm{Rad}_{\mathrm{MF}}\big(\mathrm{EL}_{12}\big(L_{\mathbb{F}_2}(1,2)\big)\big) = \mathrm{EL}_{12}\big(L_{\mathbb{F}_2}(1,2)\big),$$

which answers negatively the question of whether every countable group is MF. We also show that arbitrary countable MF groups arise as prescribed universal MF quotients while the complete MF-target representation functor is preserved.

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This manuscript is actively being revised and has not been released for public consumption.

1. Introduction

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum. Equivalently, an MF group admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements. Throughout, *MF group* means this operator-norm notion of Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]. A stronger regular-norm approximation property is also called MF in parts of the recent literature [GKEMP]; it is not the convention used here.

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For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi\colon G\to\mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

The group G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap\{\ker f : N \le \ker f,\ f\colon G \to M,\ M \text{ MF}\}.$$

Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$.

Proposition 1.1 (MF residual calculus)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If $f\colon G \to K$ is a homomorphism and $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, then every corona homomorphism of K, composed with f, kills x; hence $f(x) \in \mathrm{Rad}_{\mathrm{MF}}(K)$. This proves functoriality and, for endomorphisms, full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$ and enumerate the nonidentity elements of G/R. For each one, the definition of R supplies a corona homomorphism that detects a representative. A diagonal direct sum of coordinate models produces one operator-norm asymptotic representation that detects them all: at stage n, take the direct sum of sufficiently late coordinates of the first n models, chosen so that the first n multiplication defects are at most $1/n$ and each of the first n marked elements retains a fixed positive operator-norm separation on its designated summand. Its corona homomorphism is faithful on G/R. Thus G/R is MF.

Applied to G/N, the same argument says that G/N is MF exactly when the intersection of the kernels of all MF-target maps killing N is N, which is the asserted closure criterion. $\square$

The obstruction is captured by the following criterion.

Theorem A (one-sided compression criterion)**.** *Let G be countable, let $L \le G$ have property* (T)*, and suppose that*

$$uLu^{-1} \le L, \qquad c \in C_G(L).$$

Put

$$D = \langle\langle [ucu^{-1}, \ell] : \ell \in L \rangle\rangle_G.$$

If $K \trianglelefteq G$ has property (T) *and $K \le D$, then*

$$K \le \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF; if G has property (T) *and $D = G$, then $\mathrm{Rad}_{\mathrm{MF}}(G) = G$. Every finite-dimensional linear representation of G over every field kills D.*

One-sided Kazhdan transport puts every generator of D in the universal Hilbert–Schmidt kernel of operator-norm asymptotic representations. Normality of K makes its Kazhdan fixed-space projection invariant under the ambient group; the complementary corner detects any surviving image of K and contradicts that Hilbert–Schmidt collapse. In exact finite dimension, the corresponding commutant argument works over every field and does not require property (T).

The existence of a countable non-MF group was open [Sh]. For the binary Leavitt algebra, Theorem A answers the question. For a nontrivial simple group the MF radical is either trivial or the whole group, so full MF radical is equivalent here to failure of MF.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{1}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF.

Suppose that $L \leq G$ has property (T), that $u \in G$ satisfies $uLu^{-1} \leq L$, and that c centralizes L. For every operator-norm asymptotic representation (V_n) of G,

$$\Big\| V_n([ucu^{-1}, \ell]) - 1 \Big\|_2 \longrightarrow 0 \qquad (\ell \in L). \tag{2}$$

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the upper-left subgroup $L \cong \mathrm{EL}_3(R)$. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The element d is nontrivial and H is simple, so d normally generates H. Hence the compression defect is the ambient Kazhdan group, and Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

For an exact finite-dimensional representation, conjugation by a compressor defines an injective endomorphism of the commutant of L. This endomorphism is surjective because the commutant is finite-dimensional. Consequently, every finite-dimensional linear representation of H over every field is trivial.

The argument is specific to operator-norm approximation. It does not prove that H is nonsofic or nonhyperlinear: normalized Hilbert–Schmidt approximations do not provide the operator-norm control used to construct the conjugation representation in Theorem 3.3.

Full MF radicals determine prescribed MF quotients as follows.

Theorem C (prescribed MF quotients)**.** *Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$ and, for a countable group Q, define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q \colon W_Q \to Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q},$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

Relation to prior work. The group-MF framework used here is developed by Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]; recent permanence results are due to Shulman [Sh]. The operator-norm case remained open after non-approximability had been established for the finite Schatten norms. Bachner–Dogon–Lubotzky emphasize the interaction between operator-norm defects and Hilbert–Schmidt rigidity as a possible route to the MF problem [BDL]. Our mechanism uses that interaction differently: one-sided compression of a Kazhdan subgroup forces Hilbert–Schmidt collapse, while a normal Kazhdan subgroup inside the resulting defect converts that collapse into MF-radical membership.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's normal-subgroup theorem implies simplicity of the elementary group [Pre], while Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is finitely generated [EJZ]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho \colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. □

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator-norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator-norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator-norm asymptotic representations of G. An element is universally Hilbert–Schmidt trivial precisely when it belongs to $R_{\infty\to 2}(G)$.

Lemma 3.1. *For every sequence (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^*x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. In a stably finite unital C^*-algebra, equivalent projections $p \le q$ satisfy $p = q$. ☐

Lemma 3.2 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. ☐

Theorem 3.3 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator-norm asymptotic representation of G, and let (x_n) be uniformly bounded in operator norm. If*

$$\|V_n(\ell)x_nV_n(\ell)^* - x_n\|_2 \longrightarrow 0 \qquad (\ell \in L),$$

then, for every $\ell \in L$,

$$\|V_n(\ell)V_n(u)x_nV_n(u)^*V_n(\ell)^* - V_n(u)x_nV_n(u)^*\|_2 \longrightarrow 0,$$
$$\|V_n(\ell)V_n(u)^*x_nV_n(u)V_n(\ell)^* - V_n(u)^*x_nV_n(u)\|_2 \longrightarrow 0.$$

Proof. Let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|\,,$$

so the operator-norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$ [BHV, Chapter 3]. Since $uLu^{-1} \le L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$. Hence both U and U^* preserve $\operatorname{Fix}\sigma(L)$, so $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. The two limits in the statement therefore vanish along ω. Since this holds for every free ultrafilter, both sequences converge ordinarily. □

Corollary 3.4. *Let $L \le G$ have property* (T)*, let $uLu^{-1} \le L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, the sequence $(V_n(c))$ asymptotically commutes with $V_n(L)$ in operator norm and hence in normalized Hilbert–Schmidt norm. Theorem 3.3 shows that the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$. Therefore

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

4. Normal Kazhdan subgroups in the MF radical

Lemma 4.1 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$ and an operator-norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(q_n M_{d_n}(\mathbb{C}) q_n)$$

whose corona class is the corner representation $g \mapsto q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$. Projection lifting follows by spectral rounding. Unitary lifting follows by polar-correcting any lift, since its two unitarity defects converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. Its polar correction $W_n(g)$ is unitary and satisfies

$$\|W_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

The homomorphism relation for ρ now makes (W_n) an operator-norm asymptotic representation. Since $q \ne 0$, infinitely many q_n are nonzero; retain those coordinates. □

Theorem 4.2 (normal Kazhdan radical theorem)**.** *Let G be countable, let $D \le R_{\infty\to 2}(G)$, and let $K \trianglelefteq G$ have property* (T) *with $K \le D$. Then*

$$K \le \operatorname{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by its image and K by the image of K. The latter remains a nontrivial normal property-(T) subgroup, and every one of its elements is universally Hilbert–Schmidt trivial: any operator-norm asymptotic representation of the image composes with Θ to one of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Thus p is the projection onto the K-fixed vectors in every faithful representation of the corona. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G).$$

Put $q = 1 - p$. The projection q is nonzero, since $p = 1$ would imply that every element of K has trivial image. Lemma 4.1 gives an operator-norm asymptotic representation (W_n) on nonzero corners $q_n M_{d_n}(\mathbb{C}) q_n$.

Choose a finite symmetric Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. In the corner $q\mathcal{Q}_{\mathbf{d}}q$, the Kazhdan projection is zero. Hence the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (q\Theta(s)q - q)^*(q\Theta(s)q - q)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|}\, q.$$

Indeed, in every representation of the corner there are no K-fixed vectors, so the defining Kazhdan inequality gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - q_n)^*(W_n(s) - q_n)$$

represent b. Taking the normalized trace on $q_n M_{d_n}(\mathbb{C}) q_n$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - q_n\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

Here the Hilbert–Schmidt norms use the normalized trace of the corner. The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)q_n$ converges to zero in operator norm, which gives the displayed trace inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from the corner identity. This contradicts $s_0 \in K \leq D \leq R_{\infty\to 2}(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. Corollary 3.4 and normality of $R_{\infty\to 2}(G)$ give

$$D \leq R_{\infty\to 2}(G).$$

Theorem 4.2 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. The two stated consequences follow by taking $K \neq 1$, or $K = G = D$, respectively.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator $[ucu^{-1}, \ell]$ of D. Its kernel is normal, so it contains D. □

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (1) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{4}$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{5}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $qs_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$ proves injectivity. The term qI_3 is needed for unitality because the map $A \mapsto s_0 A t_0$ sends I_3 to pI_3.

Set

$$U = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad V = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A direct calculation using (1) gives $UV = VU = I_6$. Put

$$\tau = \mathrm{diag}(U, V) \in \mathrm{GL}_{12}(R). \tag{6}$$

Writing $I = I_6$, the Whitehead factorization is

$$\mathrm{diag}(U, U^{-1}) = \begin{pmatrix} I & U \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -U^{-1} & I \end{pmatrix} \begin{pmatrix} I & U \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}. \tag{7}$$

Each block-unipotent factor is a product of elementary 12×12 matrices: its off-diagonal 6×6 block may be inserted one entry at a time, and the corresponding matrix units have pairwise zero products. Since $V = U^{-1}$, (7) proves $\tau \in \mathrm{EL}_{12}(R)$.

Let $H = \mathrm{EL}_{12}(R)$ and let $L = \mathrm{EL}_3(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ]. Block multiplication yields

$$\tau \,\mathrm{diag}(A, I_9)\, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{8}$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0 a t_0)$. Therefore

$$\tau L \tau^{-1} \leq L. \tag{9}$$

6. The full MF radical

Proposition 6.1. *The group* $H = \mathrm{EL}_{12}(R)$ *is simple.*

Proof. The ring $R = L_{\mathbb{F}_2}(1,2)$ is purely infinite simple [AA] and therefore an exchange ring [Ara]. Let $N \trianglelefteq H$. Preusser's normal-subgroup theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R, I) \leq N \leq C_{12}(R, I),$$

where $C_{12}(R, I)$ is the full congruence subgroup of level I. Since R is simple, either $I = R$ or $I = 0$. In the first case $\mathrm{EL}_{12}(R, I) = H$, so $N = H$. In the second case

$$N \leq C_{12}(R, 0) = Z(\mathrm{GL}_{12}(R)).$$

The center of R is $\mathbb{F}_2$ by [AC, Corollary 4.3]. For completeness, suppose that $z \in Z(\mathrm{GL}_{12}(R))$. Commuting z with the elementary matrices $e_{ij}(1)$ shows that $z = \lambda I_{12}$ for some $\lambda \in R$. Commuting with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times = \mathbb{F}_2^\times = \{1\}$. Thus $Z(\mathrm{GL}_{12}(R))$ is trivial, and $N = 1$ when $I = 0$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. □

Proposition 6.2. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H*. Then*

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 3×3 corner has both indices in $\{0, 1, 2\}$. The elementary commutator relations show that it commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{10}$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$. Finally, d is nontrivial and H is simple by Proposition 6.1, so its normal closure is all of H. □

Proof of Theorem B. The group H is nontrivial and simple by Proposition 6.1. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ].

Let D be the defect subgroup in Theorem A, with $u = \tau$. The containment (9), the centrality of c relative to L, and Proposition 6.2 show that $d \in D$. Since d normally generates H, one has $D = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If M is MF and $f \colon H \to M$ is a homomorphism, compose f with a faithful homomorphism from M to a norm matrix corona. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion. □

Corollary 6.3. *For every field k and every finite-dimensional k-vector space V, every homomorphism $H \to \mathrm{GL}(V)$ is trivial.*

Proof. Let $\rho\colon H \to \mathrm{GL}(V)$. The containment (9), the centrality of c relative to L, and Theorem 2.1 give $\rho(d) = 1$. Since $H = \langle\langle d \rangle\rangle_H$ by Proposition 6.2, the homomorphism ρ is trivial. □

In particular, every homomorphism from H to a finite group, a residually finite group, or a compact Hausdorff group is trivial. For finite and residually finite targets this follows by the regular representation of a finite quotient; for compact targets it follows from Peter–Weyl point separation [PW].

Corollary 6.4 (unit groups and all matrix ranks)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$. Then the unit group $R^\times$ and $\mathrm{GL}_n(R)$ for every $n \geq 1$ are non-MF.*

Proof. For $n \geq 2$, the complete binary prefix code

$$\mathcal{C}_n = \{1^j 0 : 0 \leq j \leq n-2\} \cup \{1^{n-1}\}$$

(and $\mathcal{C}_1$ consisting of the empty word) gives a ring isomorphism

$$\Theta_n\colon M_n(R) \longrightarrow R, \qquad (a_{uv}) \longmapsto \sum_{u,v \in \mathcal{C}_n} s_u a_{uv} t_v. \tag{11}$$

Here $s_{i_1 \cdots i_r} = s_{i_1} \cdots s_{i_r}$ and $t_{i_1 \cdots i_r} = t_{i_r} \cdots t_{i_1}$. Indeed,

$$t_u s_v = \delta_{uv}, \qquad \sum_{u \in \mathcal{C}_n} s_u t_u = 1,$$

so the inverse of (11) is $r \mapsto (t_u r s_v)_{u,v}$.

Taking units gives $\mathrm{GL}_n(R) \cong R^\times$ for every n. For $n = 12$, the subgroup $H = \mathrm{EL}_{12}(R) \leq \mathrm{GL}_{12}(R)$ therefore embeds in $R^\times$. If $R^\times$ were MF, then its subgroup H would be MF, contrary to Theorem B. Hence $R^\times$ is non-MF, and the displayed isomorphisms give the result for every $\mathrm{GL}_n(R)$. □

7. Prescribed MF quotients

Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$. For a countable group Q, define

$$W_Q = B *_A (Q \times A), \tag{12}$$

where $A \to Q \times A$ is the inclusion into the second factor. The maps from B and $Q \times A$ to Q that are respectively trivial and the projection onto Q agree on A. They therefore define an epimorphism

$$\pi_Q\colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (12) are injective; in particular, $d \neq 1$ in W_Q.

Proposition 7.1 (universal factorization)**.** *Let T be a group such that every homomorphism $B \to T$ is trivial. Then precomposition with π_Q is a bijection*

$$\mathrm{Hom}(Q,T) \longrightarrow \mathrm{Hom}(W_Q,T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f\colon W_Q \to T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q,a) \in Q \times A$, we then have

$$f(q,a) = f(q,1)f(1,a) = f(q,1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills B, since $\langle\langle d \rangle\rangle_B = B$, and kills the second factor A in $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. □

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\mathrm{Rad}_{\mathrm{MF}}(B) = B$. Proposition 7.1 therefore gives the asserted bijection for every MF group M. In particular, every homomorphism from W_Q to a norm matrix corona factors uniquely through π_Q. Intersecting their kernels gives

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

□

The same factorization applies to every target group T for which all homomorphisms $B \to T$ are trivial. Taking $B = H$ and the element d from Proposition 6.2, Corollary 6.3 and Peter–Weyl point separation give the corresponding bijections for finite-dimensional linear groups over arbitrary fields, residually finite groups, and compact Hausdorff groups.

AI and computational resource usage

AI systems were used extensively for proof exploration, literature navigation, formalization, and editorial revision. Selected supporting lemmas and related variants have been formalized in Lean; this does not constitute an end-to-end formal verification of the present manuscript. The author checked and revised the mathematical arguments and citations.

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